1. The number of roots of the equation $x^3+x^2+3+2\sin x=0,\ x\epsilon[-\pi,\pi]$

NIMCET 2007

(a)2 (b)3 (c)4 (d)none

2. If $x+y=1$ and $x^3+y^3=4$, then $x^5+y^5=$

NIMCET 2007

(a)11 (b)12 (c)13 (d)22

3. Numerical value of the expression $\left|\frac{3x^3+1}{2x^2+2}\right|$ for $x=-3$ is:

NIMCET 2007

(a)4 (b)2 (c)3 (d)0

4. Let a, b and c be such that $\frac{1}{(1-x)(1-2x)(1-3x)}=\frac{a}{1-x}+\frac{b}{1-2x}+\frac{c}{1-3x}$, then $\frac{a}{1}+\frac{b}{3}+\frac{c}{5}=$

NIMCET 2007

(a)1/15 (b)1/6 (c)1/5 (d)1/3

5. If a, b are the roots of $x^2+px+1=0$ and c, d are roots of $x^2+qx+1=0$, then the value of $B=(a-c)(b-c)(a+d)(b+d)$ is

NIMCET 2008

(a)p^2-q^2 (b)q^2-p^2
(c)q^2+p^2 (d)None of these

6. Let α and β be the roots of the equation $x^2+x+1=0$. The equation whose roots are α^{19} and β^7 is

NIMCET 2008

(a)$x^2-x-1=0$ (b)$x^2+x-1=0$
(c)$x^2-x+1=0$ (d)$x^2+x+1=0$

7. The number of roots of the equation $|x^2-x-6|=x+2$ is

NIMCET 2008

(a)2 (b)3 (c)4 (d)None

8. In the expression $(x+1)(x+4)(x+9)(x+16)$ .... $(x+400)$, the coefficient of x^{19} is

NIMCET 2008

(a)2870 (b)210 (c)4001 (d)1900

9. If $2x^4+x^3-11x^2+x+2=0$, then the values of $x+\frac{1}{x}$ are

NIMCET 2009

(a)$-3,\frac{5}{2}$ (b)$-\frac{5}{2},3$ (c)$\frac{2}{5},\frac{1}{3}$ (d)$\frac{1}{3},5$

10. Find the value of k in the equation $x^3-6x^2+kx+64=0$, if it is known that the roots of the equation are n geometric progression.

NIMCET 2009

(a)24 (b)16 (c)-16 (d)-24

11. If α,β are the roots of the equation $x^2-2x+4=0$, then the value of $\alpha^6+\beta^6$ is

NIMCET 2011

(a)64 (b)128 (c)256 (d)132

12. The least integral value of k for which $(k-2)x^2+8x+k+4>0$ for all $x\,\epsilon\,R$, is

NIMCET 2011

(a)5 (b)4 (c)3 (d)6

13. The equation $(\cos p-1)x^2+(\cos p)x+\sin p=0$, where x is a variable has real roots. Then, the interval of p is

NIMCET 2012

(a)$(0,2\pi)$ (b)$(-\pi,0)$ (c)$\left(\frac{\pi}{2},-\frac{\pi}{2}\right)$ (d)$(0,\pi)$

14. Number of real roots of $3x^5+15x-8=0$ is

NIMCET 2012

(a)3 (b)5 (c)1 (d)0

15. If α and β are the roots of the equation $2x^2+2px+p^2=0$ where p is a non-zero real number and α^4 and β^4 are the roots of $x^2-rx+s=0$, then the roots of $2x^2-4p^2x+p^4-2r=0$ are

NIMCET 2014

(a)real and unequal (b) equal and zero
(c)imaginary (d) equal and non-zero

16. If $x^2-2ax+10-3a>0$ for all $x\,\epsilon\,R$, then

NIMCET 2014

(a)$-5<a<2$ (b)$a<-5$
(c)$a>5$ (d)$2<a<5$

17. The value of k for which the equation $(k-2)x^2+8x+k+4=0$, has both real, distinct and negative roots, is

NIMCET 2015

(a) 0 (b) 2 (c) 3 (d)-4

18. a, b and c are positive integrs such that $a^2+2b^2-2bc=100$ and $2ab-c^2=100$. Then, the value of $\frac{a+b}{c}$ is

NIMCET 2015

(a)10 (b)100 (c)2 (d)20

19. The value of a, for which of the sum of the squares of the roots of the equation $x^2-(a-2)x-(a+1)=0$, assumes the least value, is

NIMCET 2016

(a)3 (b)2 (c)0 (d)1

20. α,β are the roots of an equation $x^2-2x\cos\theta+1=0$, then the equation having α^n and β^n is

NIMCET 2017

(a)$x^2-(2\cos n\theta)x+1=0$
(b)$2x^2-(2\cos n\theta)x-1=0$
(c)$x^2+(2\cos n\theta)x+1=0$
(d)$x^2+(2\cos n\theta)x-1=0$

21. The equation $(x-a)^3+(x-b)^3+(x-c)^3=0$ has

NIMCET 2017

(a) all three real roots
(b) one real and two imaginary roots
(c) three real roots, namely $x=a,\ y=b,\ z=c$
(d) None of the above

22.Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\alpha/2, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of 'r' is

NIMCET 2018

(a)$\frac{2}{9}(p-q)(2q-p)$ (b)$\frac{2}{9}(q-p)(2q-p)$
(c)$\frac{2}{9}(q-2p)(2q-p)$ (d)$\frac{2}{9}(2p-q)(2q-p)$

23.The quadratic equation whose roots are $\sin^2 18°$ and $\cos^2 36°$ is

NIMCET 2018

(a)$16x^2 - 12x + 1 = 0$
(b)$16x^2 + 12x + 1 = 0$
(c)$16x^2 - 12x - 1 = 0$
(d)$16x^2 + 10x + 1 = 0$

24.Let S be the set $\{a \epsilon Z^+ : a \le 100)$. If the equation $[\tan^2 x] - \tan x - a = 0$ has real roots (where [.] is the greatest integer function), then the number of elements in S is

NIMCET 2019

(a)10 (b)8 (c)9 (d)0

25.If x is real, then the minimum value of $\frac{x^2-x+1}{x^2+x+1}$ is

NIMCET 2019

(a)1/2 (b)2 (c)3 (d)1/3

26.Let $p(x)$ be a quadratic polynomial such that $p(0) = 1$. If $p(x)$ leaves remainder 4 when divided by $x - 1$ and it leaves remainder 6 when divided by $x + 1$, then

NIMCET 2019

(a) $p(-2) = 11$ (b) $p\,(2) = 11$
(c) $p(2) = 19$ (d) $p\,(-2) = 19$

27.Number of real solutions of the equation $\sin(e^x) = 5^X + 5^{-x}$ is

NIMCET 2019

(a) 0 (b) 1
(c) 2 (d) infinitely many

28.Three cities A, B, C are equidistant from each other. A motorist travels from A to B at 30 km/h, from B to C at 40 km/h and from C to A 50 km/h. then the average speed is

NIMCET 2020

(a)39 km/h (b)40 km/h
(c)38.3 km/h (d)37.6 km/h

29.Roots of equation $ax^2 - 2bx + c = 0$ are n and m, then the value of $\frac{b}{an^2+c} + \frac{b}{am^2+c}$ is

NIMCET 2020

(a)$\frac{c}{a}$ (b)$\frac{b}{a}$ (c) $\frac{a}{c}$ (d)$\frac{b}{c}$

30.If $a + b + c = 0$, then the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$ is

NIMCET 2020

(a)1 (b)0 (c)3 (d)-1

31.For what value of p, the polynomial $x^4 - 3x^3 + 2px^2 - 6$ is exactly divisible by -1 ?

NIMCET 2021

(a)2 (b)4 (c)6 (d)8

32.If $\alpha \neq \beta$ and $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$, then the equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is

NIMCET 2021

(a)$3x^2 - 25x + 3 = 0$ (b)$3x^2 + 5x + 3 = 0$
(c)$3x^2 - 5x + 3 = 0$ (d)$3x^2 - 19x + 3 = 0$

33.If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30°$ and $\tan 15°$ respectively, then the value of $2 + P - q$ is

NIMCET 2022

(a)3 (b)0
(c)1 (d)2

34.If the equation $|x^2 - 6x + 8| = a$ has four real solution then find the value of a?

NIMCET 2023

(a)$a \,\epsilon\, 0$ (b)$a = 1$ (c)$a \,\epsilon\, (0,1)$ (d)$a \,\epsilon [1,2]$

35.If $f(x)$ is a polynomial of degree 4, $f(n) = n + 1$, n ϵ {1,2,3,4} and $f(0) = 25$, then find the value of n ?

NIMCET 2023

(a) 30 (b) 20
(c) 25 (d) None of these

36.If $|x - 6| = |x^2 - 4x| - |x^2 - 5x + 6|$, where x is a real variable.

NIMCET 2023

(a)$x = (2,5)$ (b)$x \,\epsilon [2,3] \cup [6,\infty)$
(c)$R - [2,6]$ (d)None of these

37. If α and β are the two roots of the equation $x^2 + ax + b = 0$, $(ab \neq 0)$ then the quadratic equation with roots $\frac{1}{\alpha^3+\alpha}$ and $\frac{1}{\beta^3+\beta}$ is

NIMCET 2025

(a) $b(b^2 + 1 + a^2 - 2b)x^2 + (a^3 + a - 3ab)x + 1 = 0$
(b) $b(b^2 + 1 + a^2 + 2b)x^2 + (a^3 - a - 3ab)x + 1 = 0$
(c) $b(b^2 + 1 + a^2 + 2b)x^2 - (a^3 + a - 3ab)x + 1 = 0$
(d) $b(b^2 + 1 + a^2 - 2b)x^2 - (a^3 + a - 3ab)x + 1 = 0$

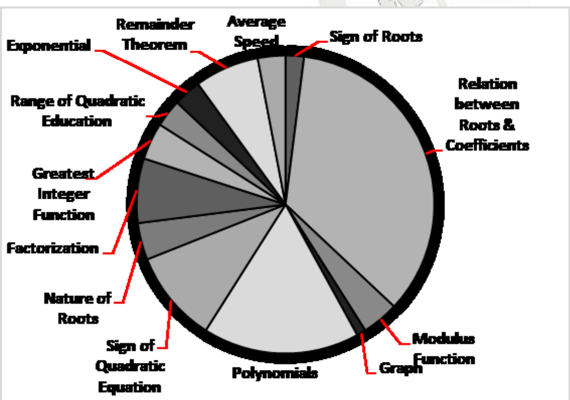
ANSWER KEY

1.	d	**2.**	a	**3.**	a	**4.**	a	**5.**	b
6.	d	**7.**	b	**8.**	a	**9.**	a	**10.**	d
11.	b	**12.**	a	**13.**	d	**14.**	c	**15.**	d
16.	a	**17.**	c	**18.**	c	**19.**	d	**20.**	a
21.	b	**22.**	d	**23.**	a	**24.**	c	**25.**	d
26.	d	**27.**	a	**28.**	c	**29.**	d	**30.**	c
31.	b	**32.**	d	**33.**	c	**34.**	c	**35.**	a
36.	b	**37.**	a						

Questions Asked in NIMCET 2007 to 2025

Years	No. of Qs.	Years	No. of Qs.
NIMCET 2007	04	NIMCET 2016	01
NIMCET 2008	04	NIMCET 2017	02
NIMCET 2009	02	NIMCET 2017	02
NIMCET 2010	00	NIMCET 2019	04
NIMCET 2011	02	NIMCET 2020	03
NIMCET 2012	02	NIMCET 2021	02
NIMCET 2013	00	NIMCET 2022	01
NIMCET 2014	02	NIMCET 2023	03
NIMCET 2015	02	NIMCET 2024	00
		NIMCET 2025	01

SUB TOPICWISE ANALYSIS



SUB TOPICWISE QUESTIONS ASKED IN NIMCET

Sub Topic	Qs. Asked	Sub topic	Qs. Asked
Relation between Root & coefficient	10	Factorization	02
Natures of Roots	01	Modulus Function	02
Sign of Roots	03	Range of Quadratic/ Quadratic	01
Sign of Quadratic Equation	03	Exponential	01
Polynomials	02	Greatest Integer Function	01
Remainer Theorem	01	Average Speed	01
Graph	01		

SOLUTION

In these type of questions, each question has four choices (a),(b),(c) and (d) of which only one option is correct.

1. (d)

2. **(a)** $x + y = 1$, $x^3 + y^3 = 4$ then $x^5 + y^5 = ?$

$x^3 + y^3 = (x+y)(x^2 + y^2 - xy)$

$4 = 1 \times ((x+y)^2 - 2xy - xy)$

$4 = 1 - 3xy$

$xy = -1$

$x^5 + y^5 = (x^3 + y^3)(x^2 + y^2) - x^2y^3 - y^2x^3$

$= (4)((x+y)^2 - 2xy) - (xy)^2(x+y)$

$= 4(1 - 2xy) - (xy)^2 . 1$

$= 4 - 8(-1) - (-1)^2$

$= 12 - 1$

$= 11$

3. **(a)** Value of $\left|\frac{3x^3+1}{2x^2+2}\right|$ for $x = -3$

$= \left|\frac{3(-3)^3 + 1}{2(-3)^2 + 2}\right|$

$= \left|\frac{-81+1}{10+2}\right| = \left|\frac{80}{20}\right| = 4$

4. **(a)**

$\frac{1}{(1-x)(1-2x)(1-3x)} = \frac{a}{1-x} + \frac{b}{1-2x} + \frac{c}{1-3x}$

For value of a

x = 1

$a = \frac{1}{(1-2)(1-3)} = \frac{1}{2}$

For value of b

X= ½

$b = \frac{1}{\left(1-\frac{1}{2}\right)\left(1-\frac{3}{2}\right)} = \frac{1}{\frac{1}{2} \times \frac{-1}{2}} = -4$

For value C

X = 1/3

$c = \frac{1}{\left(1-\frac{1}{3}\right)\left(1-\frac{2}{3}\right)} = \frac{9}{2}$

Value of $\frac{a}{1} + \frac{b}{3} + \frac{c}{5} = \frac{1}{2} + \frac{-4}{3} + \frac{9}{10}$

$= \frac{15-40+27}{30} = \frac{2}{30} = \frac{1}{15}$

5. **(b)** $a + b = -p$, $ab = 1$; $c + d = -q$; $cd = 1$

$E = (a - c)(b - c)(a + d)(b + d)$

$= [(a - c)(b + d)].[(b - c)(a + d)]$

$= [ab + ad - cb - cd].[ab + bd - ca - cd]$

$= [1 + ad - cb - 1][1 + bd - ca - 1]$

$= (ad - cb)(bd - ca)$

$= abd^2 - a^2cd - cb^2 d + c^2 ab$

Put ab = 1 and cd = 1

$= d^2 - a^2 - b^2 + c^2 = (c^2 + d^2) - (a^2 + b^2)$

$= (c^2 + d^2 + 2cd - 2cd) - (a^2 + b^2 + 2ab - 2ab)$

$= [(c + d)^2 - 2] - [(a + b)^2 - 2] = (-q)^2 - (p)^2$

$= q^2 - p^2$

It has an alternate Method:

$E = (a - c)(b - c)(a + d)(b + d)$

$= (ab - ac - bc + c^2)(ab + ad + bd + d^2)$

As we know ab = cd = 1, Hence

$E = (1 - c(a + b) + c^2)(1 + d(a + b) + d^2)$

As we know $a + b = -p$, $c + d = -q$, Hence,

$E = (1 - c(-p) + c^2)(1 + d(-p) + d^2)$

$E = (c^2 + pc + 1)(d^2 - dp + 1)$

As we know c, d are roots of $x^2 + qx + 1 = 0$, hence, c, d satisfy given equation

$c^2 + qc + 1 = 0 \Rightarrow c^2 + 1 = -qc$

and $d^2 + qd + 1 = 0 \Rightarrow d^2 + 1 = -qd$

Put these values in Eq. (i), we get

$E = (pc - qc)(-dp - qd) = -cd(p^2 - q^2)$

$= q^2 - p^2$

6. **(d)** $x^2 + x + 1 = 0$

$\Rightarrow x = \frac{-1\pm\sqrt{1-4(1)(1)}}{2(1)} = \frac{-1\pm\sqrt{3}i}{2} \Rightarrow x = \frac{-1\pm\sqrt{3}i}{2}$

$\Rightarrow x = \omega, \omega^2$

ω, ω^2 are cube roots of unity

$\omega^3 = 1$

Let $\alpha = \omega$ and $\beta = \omega^2$

Given root of Eq. to find

$\Rightarrow \alpha^{19}, \beta^7$

$\alpha^{19} = \omega^{19} = (\omega^3)^6.\ \omega = \omega.(1) = \omega = \alpha\ [\because \omega^3 = 1]$

$\beta^1 = (\omega^2) = \omega^{14} = (\omega^3)^4.\ \omega^2 = \omega^2 = \beta$

$\Rightarrow \alpha^{19} = \alpha$ and $\beta^7 = \beta$

Hence, new equation will remain same.

7. **(b)** $|x^2 - x - 6| = x + 2$

$\because |x| = a$

$\Rightarrow x = \pm a$ if $a \geq 0$

$\Rightarrow x^2 - x - 6 = \pm(x + 2)$ if $(x + 2) \geq 0 \Rightarrow x \geq -2$

$\Rightarrow x^2 - x - 6 = x + 2;\ x^2 - x - 6 = -x - 2$

$\Rightarrow x^2 - 2x - 8 = 0;\ x^2 - 4 = 0$

$\Rightarrow (x + 4)(x + 2) = 0: (x + 2)(x - 2) = 0$

$\Rightarrow x = 4, -2;\ x = 2, -2$

$\Rightarrow x = -2, 2, 4$

Hence, given equation has 3 solutions.

8. **(a)** Let $f(x) = (x + 1)(x + 4)(x + 9)\ldots\ldots\ldots(x + 400)$

$\Rightarrow (x + 1^2)(x + 2^2)(x + 3^2)\ldots\ldots\ldots(x + 20^2)$

Put f(x) = 0, we get

$X = -1^2, -2^2, -3^2, -4^2\ \ldots\ldots, -20^2$ roots of f(x)

Highest power of x in f(x) is x^{20}

Using Vieta's theorem

$f(x) = x^{20}$ – (sum of roots taken one at a time) x^{19} + (sum of root taken two at a time) x^{18} +...sum of roots taken one at a time

$= \sum\alpha = \alpha_1 + \alpha_2 + \alpha_3 + \ldots\ldots\ldots$

$= -1^2 - 2^2 = 3^2 - 4^2 - \ldots\ldots -20^2)\ x^{19} + [(-1)^2(-2)^2 + (-2)^2(-3)^2 + (-3)^2(-4)^2 + \ldots\ldots]\ x^{18} + \ldots\ldots\ldots$

Coefficient of $x^{19} = (1^2 + 2^2 + 3^2 + 4^2 + \ldots\ldots\ldots 20^2)$

$= \frac{20(21)(41)}{6}$. Put (n=20)

$= 2870$

9. **(a)** $2x^4 + x^3 - 11x^2 + x + 2 = 0$; divide by x^2

$\Rightarrow 2x^2 + x - 11 + \frac{1}{x} + \frac{2}{x^2} = 0$

$\Rightarrow 2\left(x^2 + \frac{1}{x^2}\right) + \left(x + \frac{1}{x}\right) - 11 = 0$

$\Rightarrow 2\left(\left(x + \frac{1}{x}\right)^2 - 2\right) + \left(x + \frac{1}{x}\right) - 11 = 0$

Let $\left(x + \frac{1}{x}\right) = t$

$\Rightarrow 2(t^2 - 2) + t - 11 = 0 \Rightarrow 2t^2 + t - 15 = 0$

$\Rightarrow 2t^2 + 6t - 5t - 15 = 0 \Rightarrow (2t - 5)(t + 3) = 0$

$\Rightarrow t = -3, \frac{5}{2}$

10. **(d)** Let roots be $\frac{a}{r}$, a and ar.

$\Rightarrow S_1 = \frac{a}{r} + a + ar = 6$

$S_2 = \frac{a}{r}\cdot a + a\cdot ar + ar\cdot\frac{a}{r} = k$

$S_3 = \frac{a}{r}a\cdot ar = -64 \Rightarrow a^3 = -64$

$\Rightarrow a = -4$

Put this value in Eq. (i),

$\frac{-4}{r} + (-4) + (-4r) = 6$

$\Rightarrow -4\left(\frac{1}{r} + 1 + r\right) = 6 \Rightarrow -4(1 + r + r^2) = 6r$

$\Rightarrow -4 - 4r - 4r^2 = 6r \Rightarrow 4r^2 + 10r + 4 = 0$

$\Rightarrow 2r^2 + 5r + 2 = 0 \Rightarrow 2r^2 + 4r + r + 2 = 0$

$\Rightarrow (2r + 1)(r + 2) = 0$

$\Rightarrow r = -\frac{1}{2}, -2 \Rightarrow$ taking $r = -2$

$\frac{a}{r} = \frac{-4}{-2} = 2; a = -4$

and $ar = (-4)(-2) = 8$

$\Rightarrow S_2 = \frac{a}{r}\cdot a + a(ar) + ar\cdot\frac{a}{r} = k$

$= \frac{a^2}{r} + a^2r + a^2 = a^2\left(\frac{1}{r} + r + 1\right)$

$\Rightarrow (-4)^2\left(1 + (-2) + \frac{1}{(-2)}\right) = k$

$k = -24$

11. **(b)** $\alpha + \beta = 2, \alpha\beta = 4$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 4 - 8 = -4$

$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2$

$= (-4)^2 - 2(4)^2 = 16 - 32 = -16$

$\alpha^6 + \beta^6 = (\alpha^2)^3 + (\beta^2)^3 = (\alpha^2 + \beta^2)(\alpha^4 + \beta^4 - (\alpha\beta)^2)$

$= (-4)[-16 - (4)^2] = 128$

12. **(a)** If $ax^2 + bx + c > 0 \forall x \in R$

If $a > 0$ and $D < 0$,

$(k - 2)x^2 + 8x + k + 4 > 0$ for all $x \in R$,

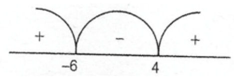
$k - 2 > 0$ and $D < 0$

$[k > 2]$ and $(8)^2 - 4(k - 2)(k + 4) < 0$

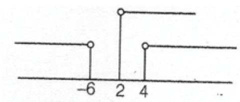
$\Rightarrow 16 - (k^2 - 8 + 2k) < 0 \Rightarrow 16 - k^2 + 8 - 2k < 0$

$\Rightarrow k^2 + 2k - 24 > 0 \Rightarrow (k + 6)(k - 4) > 0$

$\Rightarrow k < -6$ or $k > 4$



Taking intersection of these two solutions



i.e. $k > 4$ as k is an integer so least integral value of $k = 5$.

13. **(d)** $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$

coefficient of $x^2 = a \neq 0$

$\cos p - 1 \neq 0 \Rightarrow \cos p \neq 1 \Rightarrow p \neq 0, 2\pi$

Real roots ; $\Delta \geq 0 \Rightarrow \cos^2 p - 4(\cos p - 1)\sin p \geq 0$

$\cos^2 p - (4\sin p)\cos p + 4\sin p \geq 0$

Real roots ; $\Delta \geq 0 \Rightarrow \cos^2 p - 4(\cos p - 1)\sin p \geq 0$

$\Rightarrow$ This is quadratic in $\cos p$.

$\Rightarrow \cos^2 p - (4\sin p)\cos p + 4\sin p \geq 0$ to be positive always if $D \leq 0$

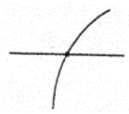
$\Rightarrow (-4\sin p)^2 - 4(1)(4\sin p) \leq 0$

$\sin^2 p - \sin p < 0 \Rightarrow \sin p(\sin p - 1) \leq 0$

as we know $-1 \leq \sin p \leq 1$ hence, $\sin p - 1 \leq 0 \forall p$

Hence, $\sin p \geq 0$
$\because \sin p \in [0,1]$
$\Rightarrow p \in [0, \pi]$
But $p \neq 0$ hence, $p \in (0, \pi]$
From given options $p \in (0, \pi)$

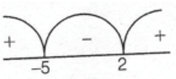
14. **(c)**. $f(x) = 3x^5 + 15x - 8 = 0$
$\Rightarrow f'(x) = 3(5x^4) + 15(1) + 0 \Rightarrow 15(x^4 + 1) > 0 \forall x \in R$
$\because f'(x) > 0 \forall x \in R$
$\Rightarrow$ $f(x)$ is an increasing function



We should know that range of odd degree polynomials are all real number.
Hence, $f(x)$ cuts at x-axis only at one point.
$\Rightarrow$ Number of real solution $= 1$

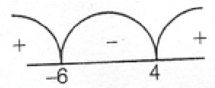
15. **(d)** $p \neq 0$ hence, $\alpha, \beta \neq 0$
$\alpha + \beta = \frac{-2p}{2} = -p$ and $\alpha\beta = \frac{p^2}{2}$
$\Rightarrow (\alpha + \beta)^2 = p^2 \Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = p^2 \Rightarrow \alpha^2 + \beta^2 = 0$
$\Rightarrow$ $\alpha^4 + \beta^4 = r$ and $(\alpha\beta)^4 = s$
$2x^2 - 4p^2x + p^4 - 2r = 0$
$D = (-4p^2)^2 - 4(2)(p^4 - 2r)$
$= 16p^4 - 8p^4 + 16r$
$= 8p^4 + 16r = 8(p^2)^2 + 16r$
$= 8(2\alpha\beta)^2 + 16(\alpha^4 + \beta^4)$
$= 32\alpha^2\beta^2 + 16(\alpha^4 + \beta^4)$
$= 16(\alpha^4 + \beta^4 + 2\alpha^2\beta^2) = 16[(\alpha^2 + \beta^2)]^2 = 0$

16. **(a)** If $ax^2 + bx + c > 0 \forall x \in R$
If $a > 0$ and $D < 0$
If $a > 0$ and $D < 0$
$D < 0 \Rightarrow (-2a)^2 - 4(1)(10 - 3a) < 0$
$\Rightarrow a^2 - 10 + 3a < 0 \Rightarrow a^2 + 3a - 10 < 0$
$\Rightarrow (a + 5)(a - 2) < 0$

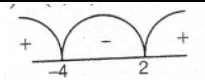


$\Rightarrow$ $a \in (-5, 2) \Rightarrow -5 < a < 2$

17. **(c)** Coefficient of x^2 cannot be equal to 0.
Hence, $(k - 2) \neq 0 \Rightarrow k \neq 2$
Condition that both roots are real, distinct and $(-)$ ve roots
$\Rightarrow D \geq 0$ and sum of roots < 0 and product of roots > 0.
Sum of roots $= \alpha + \beta = \frac{-8}{k-2} < 0$
$\Rightarrow$ $\frac{1}{k-2} > 0 \Rightarrow k > 2$
$D = (8)^2 - 4(k - 2)(k + 4) \geq 0$
$\Rightarrow$ $16 - (k - 2)(k + 4) \geq 0 \Rightarrow k^2 + 2k - 24 \leq 0$
$\Rightarrow$ $(k + 6)(k - 4) \leq 0 \Rightarrow k \in [-6, 4]$



Product of roots $= \alpha\beta = \frac{k+4}{k-2} > 0$
$\Rightarrow K \in (-\infty, -4) \cup (2, \infty)$



Taking intersection of Eqs. (i), (ii), and (iii), we get
$2 < k \leq 4$
From given options the value of $k = 3$.

18. **(C)**
$a^2 + 2b^2 - 2bc = 100$
And
$2ab - c^2 = 100$
On subtracting Eq. (ii) from Eq. (i), we get
$a^2 + 2b^2 - 2bc - 2ab + c^2 = 100 - 100$
$\Rightarrow a^2 + b^2 - 2ab + b^2 + c^2 - 2bc = 0$
$\Rightarrow (a - b)^2 + (b - c)^2 = 0 \Rightarrow a - b = 0$ and $b - c = 0$
$\Rightarrow a = b$ and $b = c \Rightarrow a = b = c = k$ (say)
$\Rightarrow \frac{a+b}{c} = \frac{2k}{k} = 2$

19. **(d):** $\alpha + \beta = a - 2; \alpha\beta = -(a + 1)$
$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (a - 2)^2 + 2(a + 1)$
$= a^2 + 4 - 4a + 2a + 2 \Rightarrow f(a) = a^2 - 2a + 6$
For least value $f'(a) = 0 \Rightarrow \frac{d}{da}(a^2 - 2a + 6) = 0$
$\Rightarrow$ $2a - 2 = 0 \Rightarrow a = 1$

20. **(a)** $\alpha + \beta = 2\cos\theta$ and $\alpha\beta = 1$
$x = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2} \Rightarrow x = \frac{2\cos\theta \pm 2\sqrt{-\sin^2\theta}}{2}$
$\Rightarrow x = \cos\theta \pm i\sin\theta$
$\Rightarrow$ $\alpha = \cos\theta + i\sin\theta; \beta = \cos\theta - i\sin\theta$
$\alpha^n = (\cos\theta + i\sin\theta)^n; \beta^n = (\cos\theta - i\sin\theta)^n$
By De-Moiver's theorem
$\alpha^n = \cos n\theta + i\sin n\theta; \beta^n = (\cos n\theta - i\sin n\theta)$
$\Rightarrow \alpha^n + \beta^n = \cos n\theta + i\sin n\theta + \cos n\theta - i\sin\theta = 2\cos n\theta$
$(\alpha\beta)^n = \alpha^n\beta^n = [(\cos n\theta + i\sin n\theta)(\cos n\theta - i\sin n\theta)]^n$
$= (1)^n = 1$
New equation $x^2 - (2\cos n\theta)x + 1 = 0$

21. **(b)** $f(x) = (x - a)^3 + (x - b)^3 + (x - c)^3 = 0$
Odd degree polynomial
$\Rightarrow$ Number of real solutions
$=$ odd i.e. possible real number of solutions $= 1$ or 3
$\Rightarrow f'(x) = 3(x - a)^2 + 3(x - b)^2 + 3(x - c)^2$
$\Rightarrow 3[(x - a)^2 + (x - b)^2 + (x - c)^2] \geq 0$
$\Rightarrow f'(x) > 0 \Rightarrow f(x)$ is an increasing function.
$\Rightarrow f(x)$ will cut the x-axis only once hence it has only one real root.
1 Real root and 2 imaginary roots.

22. **(d)** $\alpha + \beta = p; \alpha\beta = r; \frac{\alpha}{2} + 2\beta = q; \frac{\alpha}{2} \cdot 2\beta = r$
$\Rightarrow \alpha\beta = r$
$\alpha + \beta = p$
$\alpha + 4\beta = 2q$
On subtracting Eq. (ii) from Eq. (i), we get
$3\beta = 2q - p \Rightarrow \beta = \frac{(2q-p)}{3}$
$\alpha + \beta = p$
$\Rightarrow \alpha = p - \beta = p - \frac{(2q-p)}{3} = \frac{3p-2q+p}{3}$
$\Rightarrow \alpha = \frac{4p-2q}{3} = \frac{2}{3}(2p - q)$

$\Rightarrow r = \alpha = \frac{2}{3}(2p-q)\cdot\frac{1}{3}(2q-p) = \frac{2}{9}(2p-q)(2q-p)$

23. **(a)** $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ and $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$

$\Rightarrow \sin^2 18^\circ = \left(\frac{\sqrt{5}-1}{4}\right)^2$ and $\cos^2 36^\circ = \left(\frac{\sqrt{5}+1}{4}\right)^2$

$\alpha+\beta = \sin^2 18^\circ + \cos^2 36^\circ$

$= \frac{5+1-2\sqrt{5}}{16} + \frac{5+1+2\sqrt{5}}{16}$

$\alpha+\beta = \frac{6-2\sqrt{5}+6+2\sqrt{5}}{16} = \frac{12}{16}$

$\alpha\beta = \sin^2 18^\circ \cdot \cos^2 36^\circ$

$= \left(\frac{\sqrt{5}-1}{4}\right)^2 \cdot \left(\frac{\sqrt{5}+1}{4}\right)^2 = \left[\left(\frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4}\right)\right]^2$

$\alpha\beta = \left(\frac{5-1}{16}\right)^2 = \left(\frac{4}{16}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$

$\Rightarrow x^2 - (\alpha+\beta)x + \alpha\beta = 0 \Rightarrow x^2 - \frac{12}{16}x + \frac{1}{16} = 0$

$\Rightarrow 16x^2 - 12x + 1 = 0$

24. **(c)** $1 \le a \le 100$

$[\tan^2 x] - \tan x = a$ has real roots

We know that $[\tan^2 x]$ is a whole number and a is also a natural number.

Hence, $\tan x$ should also be an integer.

If any quadratic have integral roots then its discriminant must be a perfect square.

$D = b^2 - 4ac = (-1)^2 - 4(1)(-a) = 1 + 4a$

$1 + 4a$ is an odd integer

$\Rightarrow \sqrt{1+4a} = 2\lambda + 1;$

On squaring both sides, we get

$\Rightarrow 1 + 4a = 4\lambda^2 + 1 + 4\lambda \Rightarrow a = \lambda(\lambda+1)$

For different values of $a \in Z^+$and $a \le 100$

$a = 2,6,12,20,30,42,56,72,90 \Rightarrow 9$ values.

25. **(d)** $\frac{x^2-x+1}{x^2+x+1} = y$ (say) ..(i)

$\Rightarrow x^2 - x + 1 = yx^2 + xy + y$

$\Rightarrow (1-y)x^2 - (1+y)x + (1-y) = 0$

$1 - y \ne 0$; but check $\Rightarrow y = 1$ in Eq. (i)

$\Rightarrow \frac{x^2-x+1}{x^2+x+1} = 1 \Rightarrow x^2 - x + 1 = x^2 + x + 1$

$\Rightarrow x = 0 \in R; y = 1$ can be possible

As $x \in R \Rightarrow D \ge 0 \Rightarrow [-(1+y)]^2 - 4(1-y)(1-y) \ge 0$

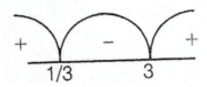
$\Rightarrow (1+y)^2 - 4(1-y)^2 \ge 0$

$\Rightarrow 1 + y^2 + 2y - 4 - 4y^2 + 8y \ge 0$

$\Rightarrow -3y^2 + 10y - 3 \ge 0 \Rightarrow 3y^2 - 10y + 3 \le 0$

$\Rightarrow 3y^2 - 9y - y + 3 \le 0$

$\Rightarrow (3y-1)(y-3) \le 0 \Rightarrow y \in \left[\frac{1}{3}, 3\right]$



$\Rightarrow$ Least value of $y = 1/3$

26. **(d)** Let $p(x) = ax^2 + bx + c = 0$

$p(0) = 1 \Rightarrow a(0)^2 + b(0) + c = 1 \Rightarrow c = 1$

By remainder theorem,

$x = 1 \Rightarrow p(1) = 4$ and at $x = -1 \Rightarrow p(-1) = 6$

$\Rightarrow p(1) = a(1)^2 + b(1) + c = 4$

$\Rightarrow a + b + c = 4$

$p(-1) = a(-1)^2 + b(-1) + c = 6$

$\Rightarrow a - b + c = 6$

Put the value of c in Eqs. (ii) and (iii), we get

$\Rightarrow +b + 1 = 4$ and $a - b + 1 = 6$

$\Rightarrow a + b = 3$ and $a - b = 5$

Solving both we get, $a = 4$ and $b = -1$

$\Rightarrow p(x) = ax^2 + bx + c = 4x^2 - x + 1$

Option (a) $p(-2) = 4(-2)^2 - (-2) + 1 = 16 + 2 + 1 = 19 \ne 11$

Hence, answer (d) is right.

27. **(a)** $\sin(e^x) = 5^x + 5^{-x} = 5^x + \frac{1}{5^x}$

$5^x > 0 \Rightarrow 5^x + \frac{1}{5^x} \ge 2 \forall x \in R$ as we know $-1 \le \sin e^x \le 1$

$\Rightarrow$ No real solutions possible.

28. **(c)** Let A, B, C are at equal distance d be x km and s_1, s_2, s_3 are speed of motorist.

$\because$ Time = Distance/Speed.

$t_1 = \frac{d}{s_1} = \frac{x}{30}; t_2 = \frac{d}{s_2} = \frac{x}{40}; t_3 = \frac{d}{s_3} = \frac{x}{50}$

$\Rightarrow$ Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

$= \frac{3d}{t_1+t_2+t_3} = \frac{3x}{\frac{x}{30}+\frac{x}{40}+\frac{x}{50}}$

$\text{LCM}(30,40,50) = 600$

$\Rightarrow \frac{3x(600)}{20x+15x+12x} = \frac{3(600)x}{47x} = 38.3$ km/h

29. **(d)** n and m are roots of $ax^2 - 2bx + c = 0$

$\Rightarrow an^2 - 2bn + c = 0$

$\Rightarrow an^2 + c = 2bn$

$am^2 - 2bm + c = 0$

$\Rightarrow am^2 + c = 2bm$

$\Rightarrow \frac{b}{an^2+c} + \frac{b}{am^2+c} = \frac{b}{2bn} + \frac{b}{2bm} = \frac{1}{2}\left[\frac{m+n}{mn}\right]$

$\because n + m = \frac{2b}{a}$ and $nm = \frac{c}{a}$

$\Rightarrow \frac{1}{2}\left[\frac{m+n}{mn}\right] \Rightarrow \frac{1}{2}\left[\frac{2b}{a}\cdot\frac{a}{c}\right] = \frac{b}{c}$

$\Rightarrow \frac{b}{an^2+c} + \frac{b}{am^2+c} = \frac{b}{c}$

30. **(c)** We know that $a^3 + b^3 + c^3 - 3abc$

$= (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$

$\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} = \frac{a^3+b^3+c^3}{abc}$

$\Rightarrow a^3 + b^3 + c^3 - 3abc = 0$

$\Rightarrow a^3 + b^3 + c^3 = 3abc$

$\Rightarrow \frac{a^3+b^3+c^3}{abc} = \frac{3abc}{abc} = 3$

31 **(b)** $\because$ divisible by $x - 1$

$\Rightarrow x - 1 = 0$ is factor of given polynomial

$\Rightarrow x = 1$ satisfy

$\Rightarrow f(1) = (1)^4 - 3(1)^3 + 2(p)(1)^2 - 6 = 0$

$\Rightarrow 1 - 3 + 2p - 6 = 0$

$\Rightarrow p = 4$

32. **(d)** $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$

$\Rightarrow \alpha^2 - 5\alpha + 3 = 0$ and $\beta^2 - 5\beta + 3 = 0$

$\Rightarrow \alpha$ and β are roots of equation

$\Rightarrow x^2 - 5x + 3 = 0$

$\Rightarrow \alpha + \beta = 5$ and $\alpha\beta = 3$

$\Rightarrow$ New Eq. has roots $= \frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

$\Rightarrow$ Sum of roots $= \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$

$= \frac{(5)^2-2(3)}{3} = \frac{25-6}{3} = \frac{19}{3}$

Product of roots $= \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$

$\Rightarrow x^2 -$ (Sum of roots) $x +$ Product of roots $= 0$

$\Rightarrow x^2 - \frac{19}{3}x + 1 = 0 \Rightarrow 3x^2 - 19x + 3 = 0$

33. **(c)** $x^2 + px + q = 0$ has roots are $\tan 30°$ and $\tan 15°$

$\tan 30° + \tan 15° = -p$

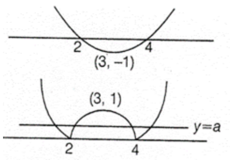
$\tan 30° \tan 15° = q$

$\tan 45° = \frac{\tan 30° + \tan 15°}{1 - \tan 30° \tan 15°} = \frac{-p}{1-q} = 1$

$-p = 1 - q \Rightarrow -1 = p - q$

$2 - 1 = 2 + p - q = 1$

34. **(c)** Let $y = x^2 - 6x + 8$



Hence, for 4 solutions a must lie between $(0,1)$.

35. **(a)** $f(x) \rightarrow 4$ degree polynomial

Let,

$f(x) = \lambda(x-1)(x-2)(x-3)(x-4) + (x+1)$

$f(0) = \lambda(-1)(-2)(-3)(-4) + 1 = 25 \Rightarrow \lambda = 1$

$f(5) = 6 + 24\lambda \qquad f(5) = 30$

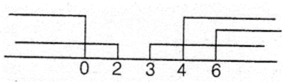
36. **(b)** $|x-6| = |x^2-4x| - |(x-3)(x-2)|$

Its one solution is

$x(x-4) \geq 0$ and $x^2 - 5x + 6 \geq 0$

And $x \geq 6$

$x \leq 0$ or $x \geq 4$ $x \leq 2$ and $x \geq 3$

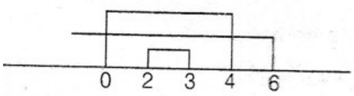


Its intersection $x \geq 6 \Rightarrow x \in [6, \infty)$

Its second solution $x(x-4) \leq 0$

and $x^2 - 5x + 6 \leq 0$ $0 \leq x \leq 4$

and $2 \leq x \leq 3$ and $x < 6$



Its intersection is $2 \leq x \leq 3 \Rightarrow x \in [2,3]$

Union of both solutions is $x \in [2,3] \cup [6, \infty)$

37. **(a)**

Given: $x^2 + ax + b = 0$, roots a, β, ab ≠ 0

We want a quadratic with roots $\frac{1}{\alpha^3+\alpha}, \frac{1}{\beta^3+\beta}$

$\alpha^2 = -a\alpha - ba^3 + \alpha^3 = \alpha(\alpha^2 + 1) = \alpha(-a\alpha - b + 1) = a^2\alpha + ab - b\alpha + \alpha = (a^2 - b + 1)x + ab$

Same for

$\beta^3 + \beta = (a^2 - b + 1)\beta + ab$

Let roots be r_1, r_2, then:

Sum $= r_1 + r_2 = \frac{1}{\alpha^3+\alpha} + \frac{1}{\beta^3+\beta}$

Product $= r_1 r_2 = \frac{1}{(\alpha^3+\alpha)(\beta^3+\beta)}$

Use identities:

$\alpha + \beta = -\alpha$, $\alpha\beta = b$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = -a^3 + 3ab$

$\alpha^3 + \beta^3 + \alpha + \beta = -a^3 + 3ab - a = -(a^3 + a - 3ab)$

$(\alpha^3 + \alpha)(\beta^3 + \beta) = b^3 + ba^2 + b - 2b^2 = b(b^2 + a^2 + 1 - 2b)$

Final quadratic:

$x^2 + (a^3 + a - 3ab)x + b(b^2 + a^2 + 1 - 2b) = 0$

Ans. $(b^2 + 1 + a^2 - 2b)x^2 + (a^3 + a - 3ab)x + 1 = 0$